Minor 1 ME150

Important Instructions

- *Each questions carries 10 marks.*
- *Write answers of every numerical on the back side of the question paper which has to be submitted indicating correct question number and sub-part number. Attach the worksheet with the question paper.*

Questions

Q1: The internal energy of a certain substance is given by the following equation.

$\boldsymbol{u = 3.56\,pv + 84}$ where u is given in kJ/kg, p is in kPa, and v is in m³/kg. A system composed of 3 kg of this substance expands from an initial pressure of 500 kPa and a volume of 0.22 m³ to a final pressure 100 kPa in a process in which pressure and volume are related by $\boldsymbol{pv^{1.2} = \text{constant}}$.

a) If the expansion is quasi-static, find Q, ΔU, and W for the process.

b) In another process the same system expands according to the same pressure–volume relationship as in part (a), and from the same initial state to the same final state as in part (a), but the heat transfer in this case is 30 kJ. Find the work transfer for this process. **[10]**

Q2: Derive efficiency of Diesel Cycle. **[10]**

Q3: A reversible heat engine operates between two thermal reservoirs maintained at temperatures of **800°C** and **60°C**. The engine drives a reversible refrigerator operating between reservoirs at **60°C** and **−30°C**. The heat supplied to the heat engine is **3000 kJ**, and the net work output of the combined engine–refrigerator plant is **600 kJ**.

Determine:

1. The work output of the heat engine.
2. The work input to the refrigerator.
3. The heat absorbed by the refrigerator from the low-temperature reservoir (−30°C).
4. The net heat transfer to the 60°C reservoir.

Assume all processes are reversible and operate on Carnot principles. **[10]**